Cracking the Code of Hallucination in LVLMs with Vision-aware Head Divergence

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Abstract

Large vision-language models (LVLMs) have made substantial progress in integrating large language models (LLMs) with visual inputs, enabling advanced multimodal reasoning. Despite their success, a persistent challenge is hallucination—where generated text fails to accurately reflect visual content—undermining both accuracy and reliability. Existing methods focus on alignment training or decoding refinements but primarily address symptoms at the generation stage without probing the underlying causes. In this work, we investigate the internal mechanisms driving hallucination in LVLMs, with an emphasis on the multi-head attention module. Specifically, we introduce Vision-aware Head Divergence (VHD), a metric that quantifies the sensitivity of attention head outputs to visual context. Based on this, our findings reveal the presence of vision-aware attention heads that are more attuned to visual information; however, the model's overreliance on its prior language patterns is closely related to hallucinations. Building on these insights, we propose Vision-aware Head Reinforcement (VHR), a training-free approach to mitigate hallucination by enhancing the role of vision-aware attention heads. Extensive experiments demonstrate that our method achieves superior performance compared to state-of-the-art approaches in mitigating hallucinations, while maintaining high efficiency with negligible additional time overhead. The code is available at https://github.com/jinghan1he/VHR.

1 Introduction

Large vision-language models (LVLMs) (Dai et al., 2023; Liu et al., 2024b) represent a notable advancement in artificial intelligence by enabling large language models (LLMs) to understand visual inputs. However, LVLMs still face the challenge of hallucination (Rohrbach et al., 2018), where generated text does not accurately correspond to visual content. This misalignment can compromise the accuracy and reliability of LVLMs across a wide range of vision and language tasks, limiting their practical applications (You et al., 2024).

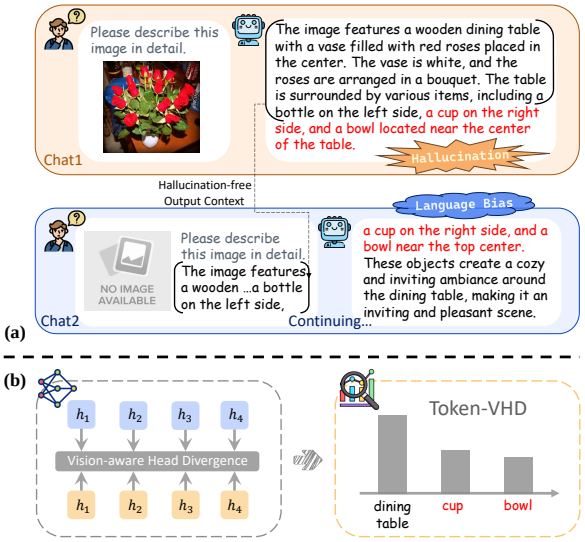


Figure 1: (a) An example indicating the connection between hallucination in LVLMs and language bias. When hallucination occurs (chat 1), we remove the image input and prompt the model to complete the description (chat 2). The output closely resembles the hallucinated content[1]. (b) The proposed VHD metric measures the sensitivity of the attention head outputs to image inputs, reflecting the degree of language bias. Hallucinated words generally correspond to lower T-VHD scores.

To mitigate this issue, several approaches incorporate additional information or models for alignment training (Zhao et al., 2023; Yu et al., 2024) or post-processing (Zhou et al., 2023; Yin et al., 2023), which incur higher training or inference costs. Recently, another line of research focuses on refining decoding strategies, employing methods like contrastive decoding (Leng et al., 2024; Zhu

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[1]Details on this example are exhibited in Appendix A.

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